



Model No.: _____
Date: _____
S/N: _____

Ice cream machine operating manual

(Please read the manual carefully before operating the machine.)

INSTALLATION NOTES

Note: In order to protect the operator from injury, the external power source of ice cream machine must be connected to the earth leakage circuit-breaker.



When installed indoors, the machine should be placed far from sprinkler heads and wetness, otherwise it can result in a severe shock .

Air-cooled machine

Please leave a space about 50cm 1.64 feet at least around the machine, this will make the air flow through the condenser adequately. If there is lack of space, refrigerating capacity will be cut down, which may cause the perpetual damage to the compressor.

Electrical connections

To install the ice cream machine in accordance with the electrical specifications of the rated power, current and voltage on the nameplate label .



Caution: This machine must be accurately grounded! Otherwise it will cause an electric shock and serious bodily injury!

When you look into the freezing cylinder, the stirring shaft must rotate in a clockwise direction.



Note: The following steps should be carried out by professional repair technicians.

When phase reversal happens on power supply, two of the live wires should be exchanged to install.

SAFETY MEASURES



Important note: Violating the safety regulations may result in heavy casualties. Failure to comply with this warning will damage the machine and its components. Component damage will increase depending on the spare parts and maintenance.

Safe operation:



Do not operate the ice cream machine before read the instructions carefully. Failure to follow this instruction will cause damage to the machine or operator.



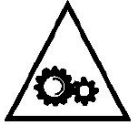
- Do not operate the ice cream machine that is improperly grounded .

- Do not repair the ice cream machine before disconnect the main power, otherwise it will cause an electric shock. Please contact with local agent or find a professional repair personnel when necessary.

Failure to follow these instructions could result in electrical shock or damage to the machine, please consult the electrician for the correct operation when you are confused .



The machine must be placed on horizontal plane, otherwise it may cause injury to persons or damage to machine.



- Do not allow untrained persons to operate the machine.
- Before disconnect the power supply, do not disassemble the throat block, the agitating shaft or transmission shaft.
- Do not dive objects or fingers into the throat block discharge outlet.

NOTE: Failure to follow these instructions can lead to serious injury because of dangerous moving parts.



Be careful when you disassemble the agitating shaft (disconnect the power supply before the operation), otherwise it may cause injury.

PRECAUTION

●Precaution before using the ice cream machine:

1. The machine is inevitably subject to shocks in the transportation and it should be used only after 24 hours at best.
2. There are different environment temperature, and it's necessary to avoid the sun and be away from heat.
3. It is suggested to use the slurry at the 25°C.
4. The machine should be placed in good ventilated location and on horizontal position.
5. The rotation direction of the agitating shaft should be in clockwise when you face to the machine.
6. The air inlet of the machine (left / right / back board) must be maintained proper ventilation.
7. It's best to clear the dirt on the condenser once a month.
8. Notices for air-cooled machine:

Leaving a space about 50cm at least around the machine can make air move in. The machine under part can not place anything, which can ensure cooling.

9. The machine can not work when it's short of material in the hopper.
10. Please disconnect the power supply when the machine is not used for a long time.

●Notices for the power line:

- a) Power socket must be equipped with ground wire, or the ground wire must be connected the connector which is near the power line and on the back of the machine.
- b) The cross-sectional area of power line must be larger than 6 square millimeters(or

10 AWG) when the machine is plugged, which can ensure the quantity of electricity in the circuit.

Note:

1. If the current capacity of the main power cord becomes low or overload, it will drop the voltage. Then the machine can not be used normally and even becomes damaged.
2. If the power cord need to be replaced, you have to find the professional maintenance personnel to replace the power cord that meets the criteria.

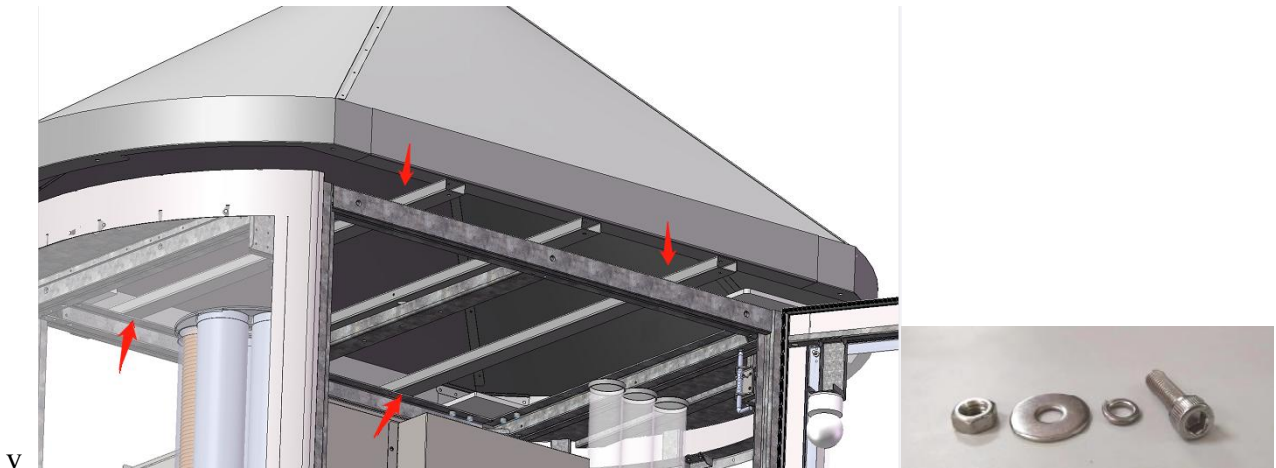
Warm tips:

The machine is designed to work normally indoors with the normal temperature(between 21°C and 28 °C).

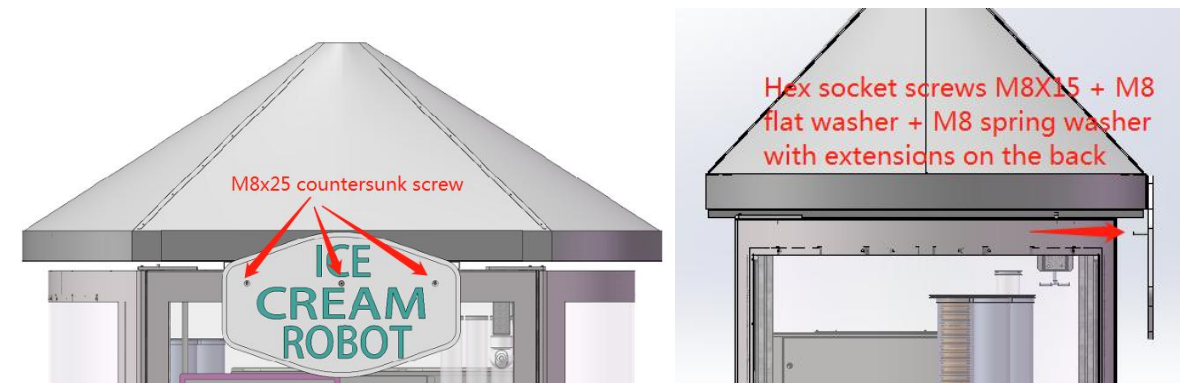
When the temperature is higher than or equal to 32°C, the production output may reduce and the ice cream may be abnormal.

INSTALLATION

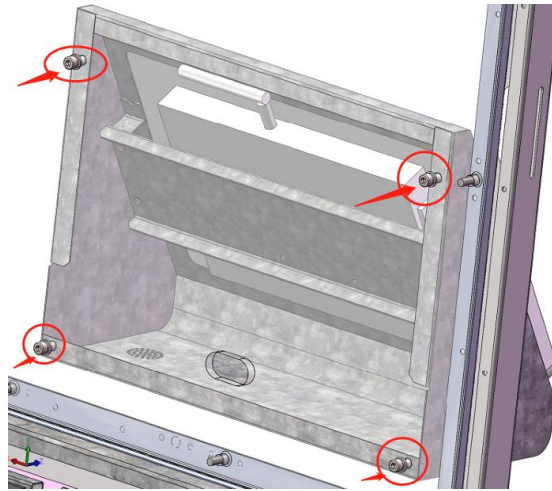
1. Roof top - screw to the rack by four M6X20 hexagon socket screw, M6 nuts, M6 flat washer and M6 spring washer.



2. Billboard - fix the billboard to the front door with three countersunk head cross screw M8X30 on the front and three hexagon socket screws M8X15 + M8 flat washer + M8 spring washer with extensions on the back.



3. Front box for tablet fixed with four M6X20 hexagon socket screw, M6 flat washer and M6 spring washer



Before using the machine for the first time (as well as for daily maintenance), perform the following steps.

For this you need:

- At least 2 buckets of lukewarm water;
- Disinfectants and detergents;
- Cleaning brushes and lubricants;
- Clean towel.

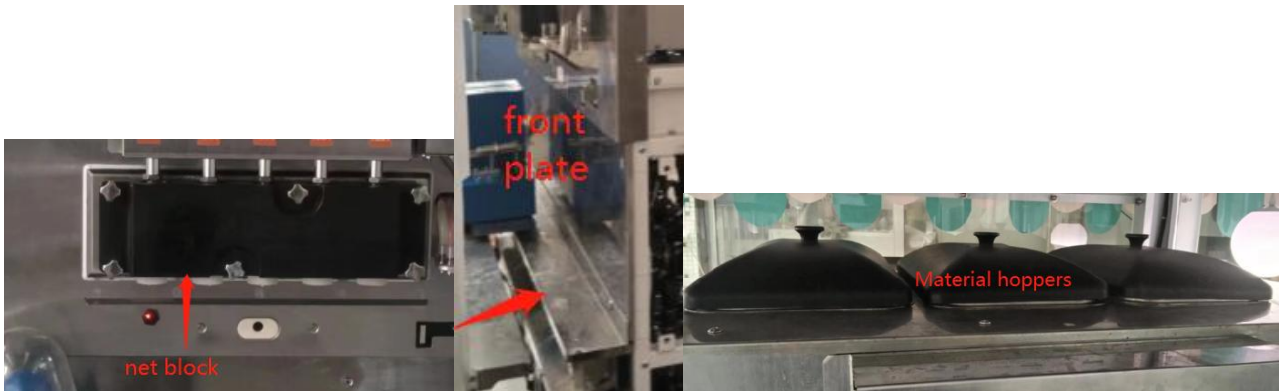
Step 1: Prepare an appropriate disinfectant or cleaning liquid. Use lukewarm water when cleaning.

Step 2: Pour the sanitizer into the hoppers, press the pump button  to pump the sanitizer into cylinder until there is liquid come out from the overflow valve, the pump will stop working after finish pumping, then press the Mixing button  and run the agitator shaft for about 5 minutes. Press the Mixing button  again to finish the process. Then drain dirty liquid from the machine.



Step 3: Now pour clean water into the hoppers (recommended water temperature about 45-50°C), press the pump button  to pump the water into cylinder until there is water come out from the overflow valve, the pump will stop working after finish pumping, then press the Mixing button  and run the agitator shaft for about 3 minutes. Then press the Mixing button  again to stop. This step can be repeated several times if necessary. After that, remove the dirty liquid from the machine.

Step 4: With a clean towel, clean the neck block, front plate and hoppers.



Step 5: To resume operation, restart the machine. Please check if any error messages are displayed on the control panel before proceeding with operation.

For error messages on the control panel, see TROUBLESHOOTING.

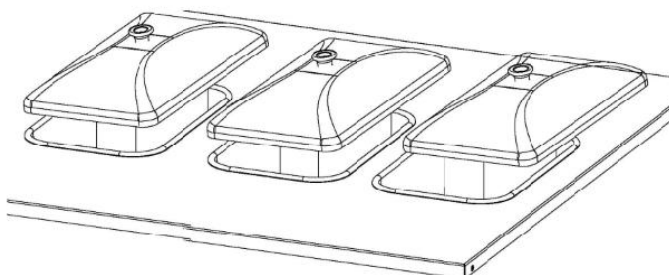
START ICE PRODUCTION



CAUTION! The machine is inevitably exposed to vibrations during transport. It should therefore only be put into operation after 24 hours.

Step 1: Prepare the ice cream mixture and pour it into the hoppers.

Step 2: Cover the hoppers with the lid (pictured below).

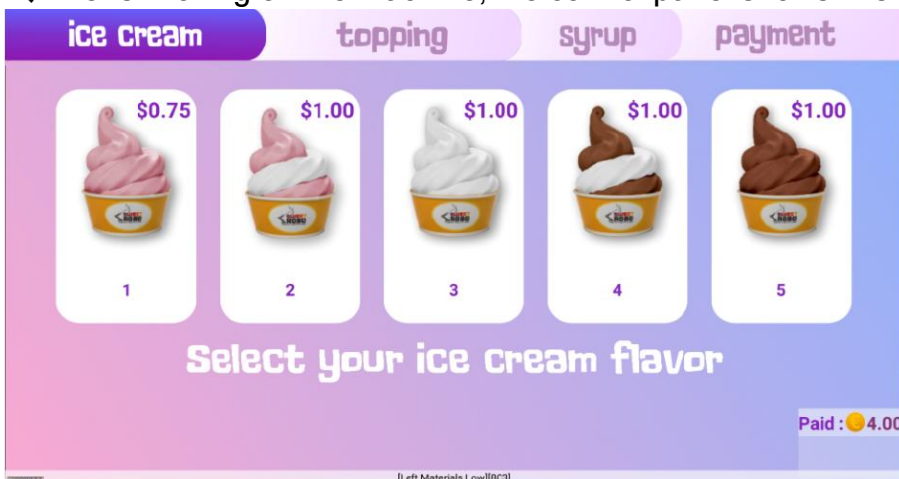


Step 3: press the pump button  to allow the ice cream mix to flow well into the freeze cylinder until there is mixture come out from the overflow valve, the pump will stop working after that, press the freeze button  to produce soft serve.

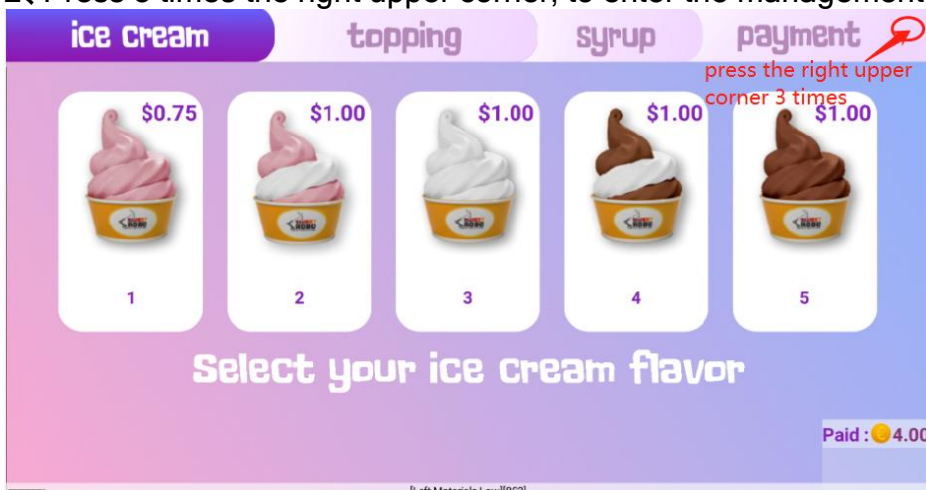


Control Panel

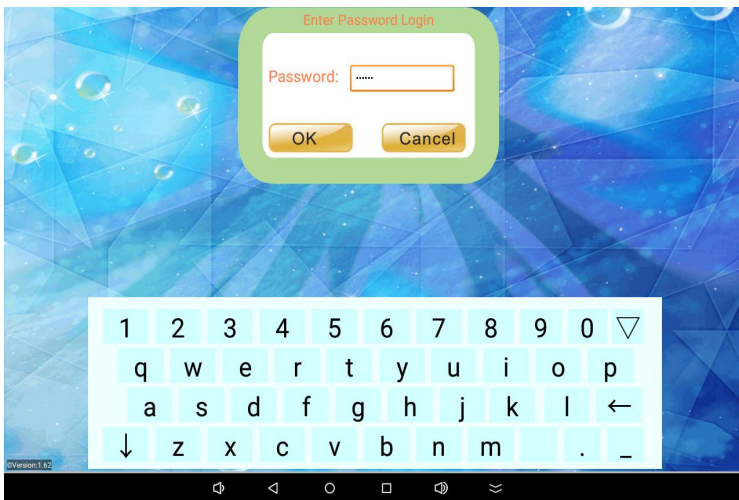
1. After switching on the machine, the control panel shows this interface:



2. Press 3 times the right upper corner, to enter the management background



3. Following screen will pop up



The preset passwords are "admin" and "user".

The password "admin" has the authorization to call up the following functions:

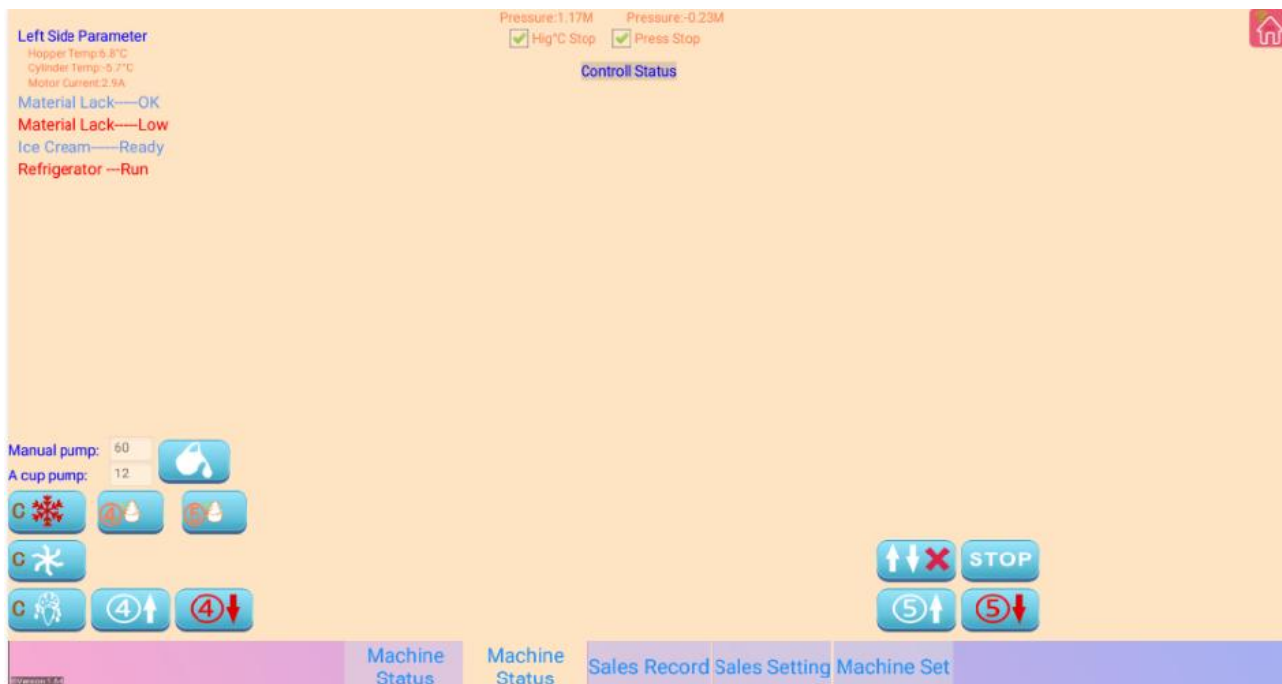
- machine setting;
- Sales setting;
- Machine status;
- Sales data

The password "user" can only call up the machine status and sales data.

After entering the "OK" password, press "OK" to get into the respective menu. The various administrative and setting options are now explained:

4. Press "Machine Status" to manage the functions and processes of the machine.





Key	Feature Instruction	Key	Feature Instruction
	Produce ice cream, pre-cooling for A hopper		A Cylinder mixing
	Produce ice cream, pre-cooling for B hopper		B Cylinder mixing
	Produce ice cream, pre-cooling for C hopper		C Cylinder mixing
	A cylinder Defrost		Heat treatment (Pasteurization)
	B cylinder Defrost		Keep fresh and pre-cooling
	C cylinder Defrost		Stop all failure and cooling system
	Lift up the first plunger tappet		Drop down the first plunger tappet
	Lift up the 2nd plunger tappet		Drop down the 2nd plunger tappet
	Lift up the 3rd plunger tappet		Drop down the 3rd plunger tappet
	Lift up the 4th plunger tappet		Drop down the 4th plunger tappet
	Lift up the 5th plunger tappet		Drop down the 5th plunger tappet
	Stop tappet running, reset the cup error		Cup holder rotation
	Drop the cup		Reset key for get back action (When prompted coin shortage, added after the

USER MANUAL

			coin must press this button)
	ON-OFF switch for Automatic door		Air pump
	Topping 1		Syrup 1
	Topping 2		Syrup 2
	Topping 3		Syrup 3
	Lift up the Z shaft		Move left the X shaft
	Drop down the Z shaft		Move right the X shaft
	Test for the 1 st position		Test for the 4 th position
	Test for the 2 nd position		Test for the 5 th position
	Test for the 3 rd position		

Press button  to run the peristaltic pump, and the icon becomes red color; when the pump finished working, press button  to run the pre-cooling procedure of material hopper and produce ice cream,

When it displays Ice Cream - Ready, then the ice cream is ok to sell ;

Press key  to run the cylinder mixing procedure, it will stop automatically after 5 minutes, and also can press the key to stop during the procedure at any time.

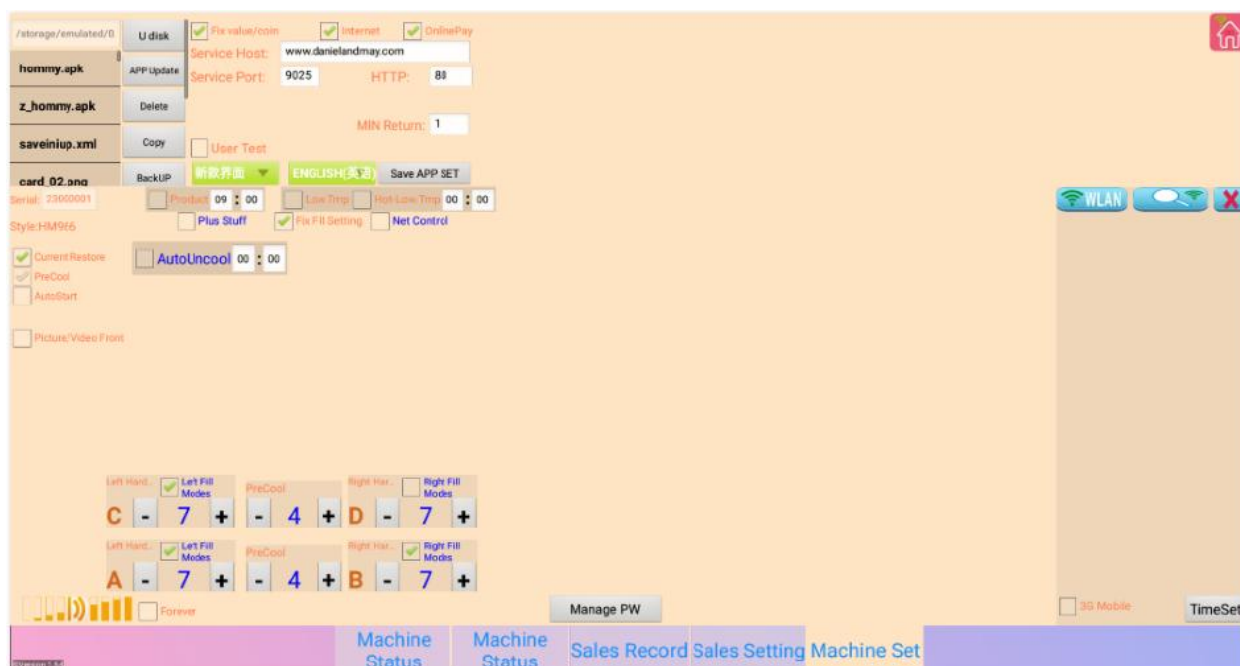
Press key  to run the cylinder defrost procedure, it will stop automatically when finishes, and also can press the key to stop during the procedure at any time. The function can defrost the ice cream in cylinder, to drain out and keep the material at fridge for next day using.

Press down the key  to run the pre-cooling and keep fresh procedure, the temperature of the material in hopper will be dropped down in a certain time.

5. Press "Sales Setting" to set the weight of the ice cream portion and the price. You can also set various payment methods and other properties



6. Press “Machine Set” to set hardness of the ice cream and the time of of the pre -cool mode and Pasteurization and AutoUncool. Here you can also activate the WLAN connection of the machine and connect the machine to a WLAN. and Click on the icon  is to adjust the voice volume



Remarks:

- 1) Mark "Hot-Low Tmp" and set the time 00:00 for example, then the machine will start pasteurization automatically at 0:00 every day. The pasteurization takes around 2~2.5 hours. After finished pasteurization at 2:00~2:30, the machine will be under standby mode and keep the ice cream in 4~5°C.
 - 2) Mark "Product" and set the time 07:00 for example, then the machine will stop standby mode and start freezing automatically at 07:00.
- (***if only marked "Hot-Low Tmp" but didn't mark "Product", then the cylinder can't start freezing automatically, and the machine will always be under standby mode)
- 3) Mark "AutoUncool" and no need to set the time, then the machine will automatically

defrost the ice cream in cylinder into liquid mixture when there is no order in around 2 hours, and it will reproduce the liquid mixture into ice cream automatically, to make sure no ice crystal created in the cylinder and avoid frozen cylinder problem. When it is set the time 01:00 for example, then the ice cream will be defrost only one time at the time 01.00

4) Mark "AutoStart", the machine will restart automatically if it is power cut not more than one hour.

7. Press Sales Record ("sales") to view sales and business volume.

Serial

PAY Type

ICE Place

ICE Cream

ICE Price

Weight(g)

Top Ping

Topping Price

Coins Cash

Sales

Subtotal

Balance

Card.No

Pay.no/Discount

Ex_Date

Total: 0

Cash: 0

Token: 0

WeChat: 0

Alipay: 0

Page: 0

Records: 0

Test: 0

Err: 0

Net.SN 0

Qr.Sale 0

Sales 0

2023-02-28

2023-02-28

00:00

00:00

SALE

Page UP

Page Down

ReSend

Query Net

Logs

DB Backup

Machine Status

Machine Status

Sales Record

Sales Setting

Machine Set

Serial	PAY Type	ICE Place	ICE Cream	ICE Price	Weight(g)	Top Ping	Topping Price	Coins Cash	Sales	Subtotal	Balance	Card.No	Pay.no/Discount	Ex_Date
129	Card.1	MM	5	1.00	0	0	0.00	0.00	0.00	1.00	0.00	23000001_M0_202302		2023-02-27 19:14:51
128	Card.1	L	1	0.75	0	0	0.00	0.00	0.00	0.75	0.00	23000001_M0_202302		2023-02-27 19:13:23
127	Cash	L	1	0.75	0	0	0.00	0.75	0.00	0.75	14.00	23000001_CS_2023022	0	2023-02-27 18:54:29
126	Cash	MM	5	1.00	0	0	0.00	1.00	0.00	1.00	15.00	23000001_CS_2023022	0	2023-02-27 18:29:39
125	Cash	RR	3	1.00	0	0	0.00	1.00	0.00	1.00	1.00	23000001_CS_2023022	0	2023-02-27 18:25:08
124	Cash	MM	5	1.00	0	[2]	0.00	1.00	0.00	1.00	2.25	23000001_CS_2023022	0	2023-02-27 18:15:03
123	Cash	L	1	0.75	0	0	0.00	0.75	0.00	0.75	3.00	23000001_CS_2023022	0	2023-02-27 18:13:47
122	Cash	RR	3	1.00	0	0	0.00	1.00	0.00	1.00	4.00	23000001_CS_2023022	0	2023-02-27 18:12:27
121	Cash	MM	5	1.00	0	0	0.00	1.00	0.00	1.00	5.00	23000001_CS_2023022	0	2023-02-27 18:11:12
120	Cash	RR	3	1.00	0	0	0.00	1.00	0.00	1.00	6.00	23000001_CS_2023022	0	2023-02-27 18:10:02
119	Cash	RR	3	1.00	0	0	0.00	1.00	0.00	1.00	7.00	23000001_CS_2023022	0	2023-02-27 18:08:16
118	Cash	L	1	0.75	0	0	0.00	0.75	0.00	0.75	7.75	23000001_CS_2023022	0	2023-02-27 18:05:29
117	Cash	MM	5	1.00	0	0	0.00	1.00	0.00	1.00	8.75	23000001_CS_2023022	0	2023-02-27 18:03:43
116	Cash	MM	5	1.00	0	0	0.00	1.00	0.00	1.00	9.75	23000001_CS_2023022	0	2023-02-27 18:02:25
115	Cash	MM	5	1.00	0	0	0.00	1.00	0.00	1.00	10.75	23000001_CS_2023022	0	2023-02-27 18:01:07
114	Cash	MM	5	1.00	0	0	0.00	1.00	0.00	1.00	11.75	23000001_CS_2023022	0	2023-02-27 17:59:39
113	Cash	L	1	0.75	0	0	0.00	0.75	0.00	0.75	12.50	23000001_CS_2023022	0	2023-02-27 17:56:30
112	Cash	MM	5	1.00	0	0	0.00	1.00	0.00	1.00	13.50	23000001_CS_2023022	0	2023-02-27 17:55:02

Total: 46.75

Cash: 35.0

Token: 0.0

WeChat: 0.0

Alipay: 0.0

Page: 1/3

Records: 51

Test: 2.5

Err: 0.0

Net.SN 0.0

Qr.Sale 0.0

Sales 0.0

2023-02-27

2023-02-28

00:00

00:00

SALE

Page UP

Page Down

ReSend

Query Net

Logs

DB Backup

Machine Status

Machine Status

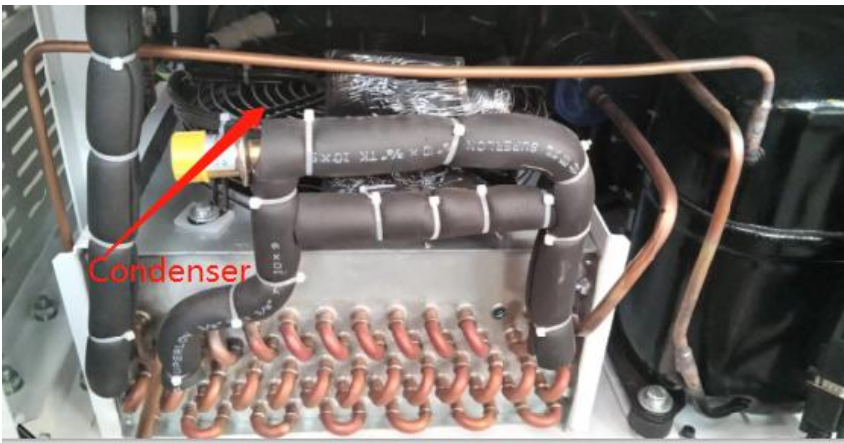
Sales Record

Sales Setting

Machine Set

Routine maintenance

1. Control of the seals: check all seals regularly and replace all damaged seals. All seals must be in good condition so that the Machine works perfectly. Replace defective seals immediately! Normally we suggest to replace all the gaskets and the plastic scrapers inside the cylinder every 3 months.
2. For Air cooling system machine, check the condenser: regularly check whether the condenser has accumulated dust and fluff. Dirty condenser will reduce the efficiency and production capacity of the machine. Clean the condenser with a soft brush every month. Do not use metal items to clean the ribs.



ATTENTION! When working on the condenser, power supply must be Turned off. Otherwise there is a chance of an electric shock danger!

3. For the water cooling system machine, check whether there is leakage or kink on the soft water pipe. With the cleaning and maintenance of machines when moving back and forth may cause a kink, if the soft water pipe is damaged or broken should only be replaced by professional staff.
4. The motors should be oiled every 6 months, to reduce wear and maintain flexible operation.

MACHINE CLEANING, MAINTENANCE AND USAGE

●Machine operating considerations and relevant job program

1. The need for machine cleaning at the end of each day and the next day the first use of the ice cream machine, ensure that the entire machine clean.

2. The cleaning process, should find it possible or damaged parts should be replaced as soon as possible; find other may result in damage or other adverse circumstances, require early treatment. If an unknown contact after-sales consultation, or seek professional help.
3. It is important to clean the machine, but do not ignore hygiene business premises surrounding environment.
4. Note if the ice cream machine environment meet the requirements.
5. Cleaning and disinfection process, the following points should be stressed.
 - 1) In the cleaning process, some steps still need the machine powered, do not need outside of the liquid splashing into cleaning position place to prevent accidents;
 - 2) Clean the body of the machine and during the ice cream machine is powered prohibit manipulation of two or more control panels, or they may cause harm to others;
 - 3) Machine regularly thorough cleaning and disinfection, includes complete disassembly of the brush and cleaning.
 - 4) After each cleaning, the used brushes, towels and other cleaning utensils should be cleaned thoroughly and disinfected.
 - 5) Thoroughly clean the mixing tank is located behind the rear seat assembly bearing and the throat.



- 6) Properly prepare the cleaning and sanitizing solutions, carefully read and follow the instructions on the label. Too strong of a solution may damage the parts and too weak cleaning solution will not provide adequate cleaning or disinfection capabilities.
- 7) After cleaning and disinfection) must be used up per day, subject to simple cleaning

●Preparation and cleaning utensils:

- 1) Two or more the number of buckets (required temperature).
- 2) Special disinfectant or cleaning fluid.
- 3) Accessories - brush package brush and food grade grease.
- 4) Cleaning cloth towels and other items.
- 5) Ensure a clean place vacant for temporarily placing clean parts.
- 6) A Phillips screwdriver, Allen key 1 set.

Note: In cleaning process, careful attention to the motor or other devices charged, if necessary, consult the relevant professionals auxiliary.

●Cleaning machine Instruction steps:

Step 1: stop the machine and leave it on standby, remove the air pump and overflow value.



step 2: Press down button  to do defrost process, after a few minutes, when the material in cylinder melts, press down button  again to stop. Press down button  to dispense out the material or liquid from the throat block, after that, press down the button  again to stop.

Step 3 : install the air pump component.

Step 4: Pour clean warm water into 2/3 hopper (requires the water temperature is 45~50℃), press down the key  to run the peristaltic pump (this operating is cleaning the piping inside the peristaltic pump), then press down the key  to run the stirring shaft for about 3 minutes, press the key again to stop. This step should be carried on for 1-2 times and then remove the air pump component.

Step 5: Place a bucket under the throat block outlet, or connect a hose, operate the keyboard to lift up the tappets and the plungers, press down button  to discharge out the dirty water to outside. After that, stops all operations, keep the machine at stand-by mode. (Note: Be careful to scald the hand that the water is still in high temperature.)

Step 6: Prepare one bucket of warm water (the water temperature is about 50℃ or above), then remove and take away the mixer upper cover, place them in the warm water.

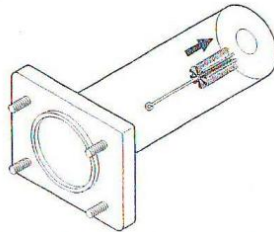
Step 7: Operate the keyboard to lift up the tappets and the plungers, then remove the throat block, and pull out the stirring shaft, the transmission shaft, the silicone gasket from the freezing cylinder one by one, then place them into the bucket with clean warm water. After that, turn off the machine power.

Step 8: Prepare qualified disinfectant or cleaning liquid. Use warm water and follow the manufacturer's specifications.

Step 9: remove all the components in the warm water, clean all the parts. Then install all the parts and place them on the clean and dry bucket.

Step 10: Ensure the machine has been stopped running, clean the hopper and cylinder inside with cleaning liquid, brushes and towel.

After cleaning, clean the components again in clean warm water to remove the remainder cleaning liquid, then wipe and dry the components with clean towel



Step 11: Clean the cone tray, the area around the throat block and the hopper with cleaning liquid.

Step 12: step 1 to 10 are finished, install all the parts in order.

Turn on the power and make the piston pull up ➡ install the transmission shaft and stirring shaft ➡ install the throat block ➡ install air pump component ➡ install the mixer ➡ cover the lid of hopper

Note: when you install the air pump component, you need to cover lube on the gasket position.

● Machine maintenance

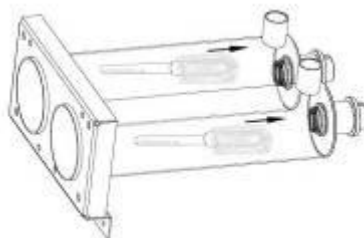
1、Cleaning with brushes

Step 1 : Preparation of qualified cleaner / disinfectant. Use warm water and follow the manufacturer's specifications.

Step 2: To thoroughly clean all disassembled parts with brushes in disinfectant and clean liquid, be sure to remove all of the oil and the slurry film. Be sure to clean the surface of all parts with brushes.

Clean all the parts with clean warm water, place these parts on a clean surface and dry all.

Step 3: Take a little clean liquid into cylinder, clean the rear bearing sleeve behind the cylinder with brushes, as shown as below figure.



Step 4 : Clean and wipe all the outer surface of ice cream machine with a clean and

disinfected towel.

2、In cleaning and disinfecting

Cleaning and disinfecting schedules should be managed by your local management, and must strictly abide. In the cleaning and disinfecting process, should emphasize the following points.

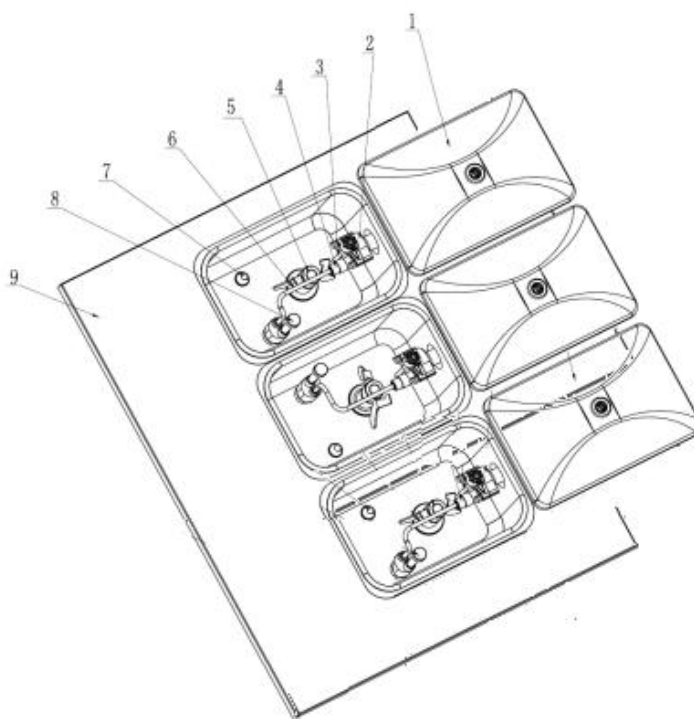


It must clean and disinfect every week.

1. Regular thorough cleaning and disinfecting machine, including completely remove and clean with the brushes.
2. All brushes provide a thorough cleaning.
3. Thorough cleaning the rear bearing sleeve behind the cylinder, and the air pump assembly and the throat block assembly.
4. The right to cleaning and disinfection liquid, carefully read and follow the instructions on the label. The cleaning liquid is too strong may damage the parts, cleaning liquid is too weak to be unable to provide adequate cleaning or disinfecting.

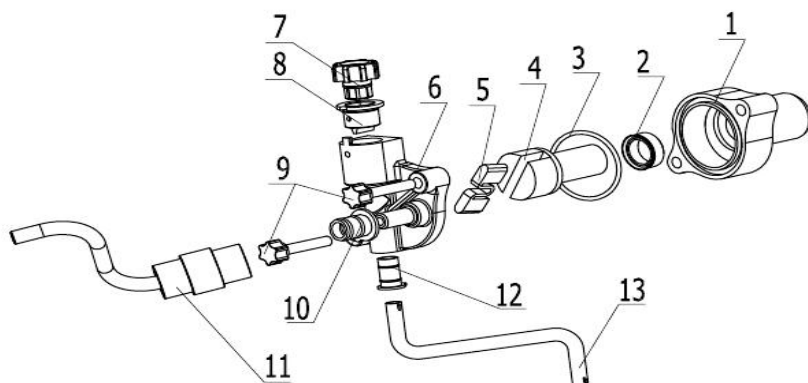
ASSEMBLY AND DISASSEMBLY OF INDIVIDUAL PARTS

1. Top plate



Ref. No.	Description	Quantity
1	Hopper lid	1
2	Air pump	1
3	Feeding tube	1
4	Connection plug	1
5	Mixer	1
6	Joint pipe	1
7	Liquid-level sensor	1
8	Overflow valve	1
9	Top board	1

2. Air pump

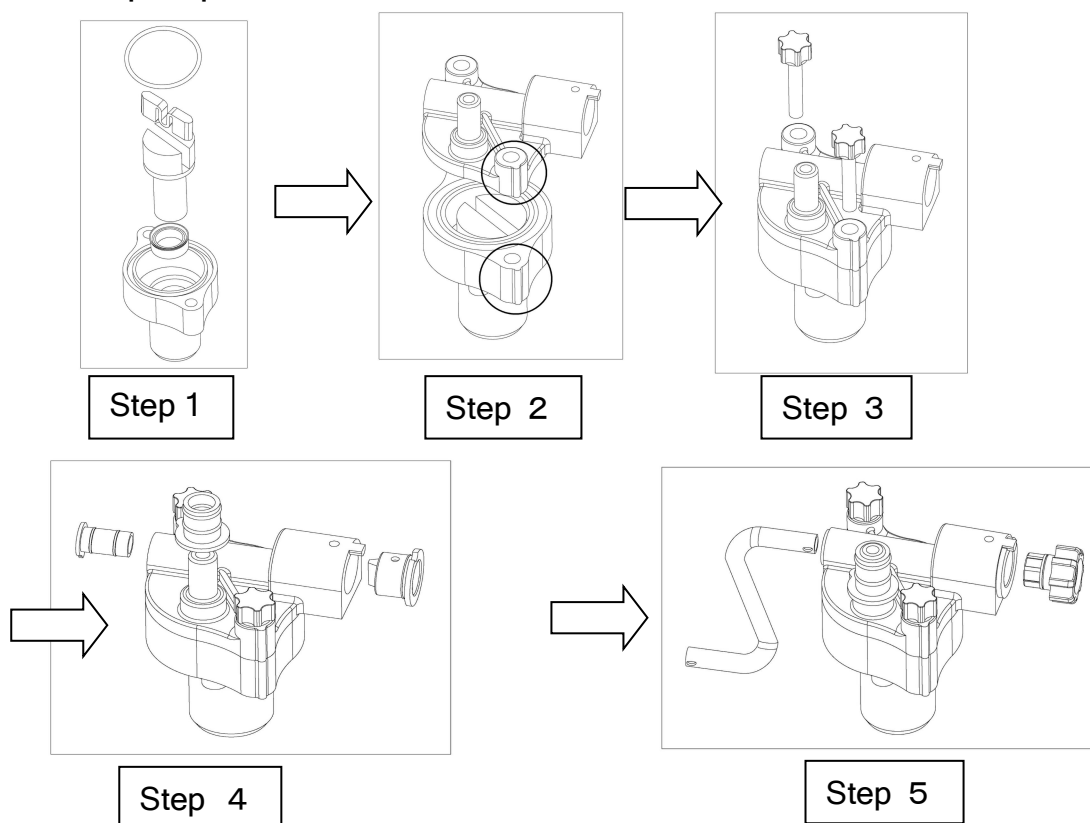


Ref. No.	Description	Quantity	Ref. No.	Description	Quantity
1	Pump body	1	8	Check value	1
2	Bearing sealφ16	1	9	Fixing bolt	2
3	Sealing ringφ50×φ3	1	10	Outlet sealing ring	1
4	Rotary axis	1	11	Joint pipe	1
5	Blade	1	12	Sealing ring of feeding tube	1
6	Pump cover	1	13	Feeding tube	1
7	Air regulator	1			

Note:

- 1) The expansion rate is changed through the blowhole of air regulator. There are eight gears to adjust the expansion rate, and it's lower in turn from 1st gear to 8th gear.
- 2) The position of sealing rings should be covered with lube in re-installation after cleaning.

Air pump installation



All of the sealing rings must be in good condition and then the machine work normally. If any cracks on the sealing rings and sealing gaskets, please replace them at once and throw away the damaged spare parts.

Note: all of the spare parts must be disinfected after each cleaning.

The steps are following:

Step 1: Install the bearing seal, sealing ring $\phi 50 \times \phi 3$, blade, rotary axis and sealing ring of feeding tube.

Note: the position of sealing rings must be cover with lube.

- ① Press the bearing seal into the air pump body.
- ② Press the rotary axis into the air pump body.
- ③ Insert the blade into the rotary axis.
- ④ Insert the sealing ring($\phi 50 \times \phi 3$) into the groove of air pump body.

Step 2: Air pump cover installation:the cover should be placed flat on the air pump body.

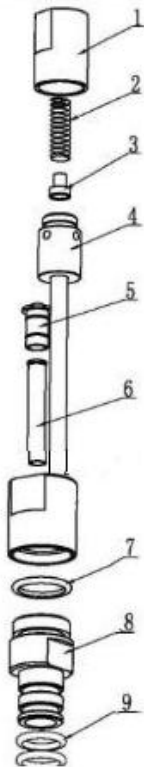
Note:The installation must ensure that the strengthened ribs on the top position and on the pump body must overlap. Every time you install the air pump, you should coat the lube on the surface inside the air pump to reduce the friction.

Step 3: Install and tighten the air pump cover and body with the fixing bolts.

Step 4: Install the check valve, outlet sealing ring and sealing gasket of feeding tube.

Step 5: Install the air regulator and feeding tube.

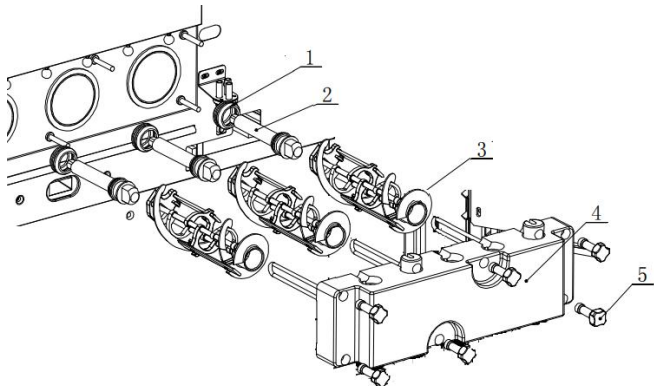
3. Overflow value



Ref. No.	Description	Quantity
1	Adjusting nut	1
2	Spring	1
3	Sealing gasket	1
4	Overflow value pedestal	1
5	Sealing plug	1
6	Blanking tube	1
7	Value seat	1
8	O-ring	1
9	Blanking set	1
10	O-ring	2

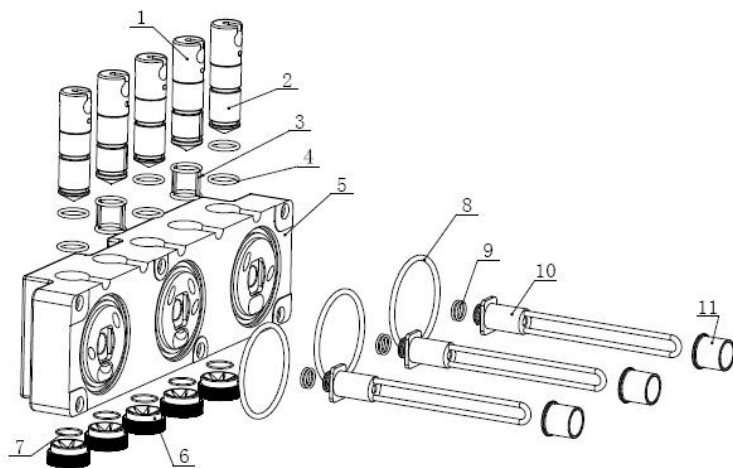
4. Stirring shaft

When you install the stirring shaft, two orifices of plastic scrapper should be positioned to the locks and pressed in. (Note: the sharp side of the plastic scrapper must be outward.) It can not be installed if sheath A and sheath B can not be exchanged. And the locating sleeve should be installed into the shaft core.



Ref. No.	Description	Quantity
1	Gasket	3
2	Transmission shaft	3
3	Stirring shaft	3
4	Throat block	1
5	Hand screw	6

5. Throat block module

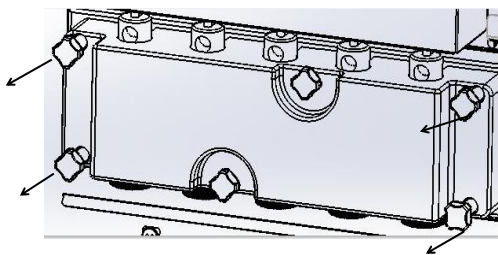


Ref. No.	Description	Quantity
1	Middle plunger	2
2	Side plunger	3
3	Bridge gasket $\Phi 33.5$ for middle plunger	2
4	Gasket $\Phi 34$ for side plunger	6
5	Throat block	1
6	Star shape lid	5
5	Middle piston	1
6	Bridge gasket	1
7	O shape gasket $\Phi 29$	5
8	O shape gasket $\Phi 105$	3
9	O shape gasket $\Phi 22$	6

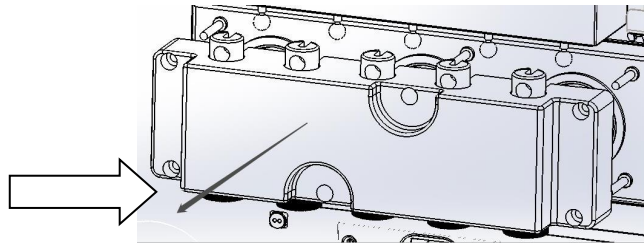
10	U shape shaft core	3
11	Locating sleeve	3

Note:

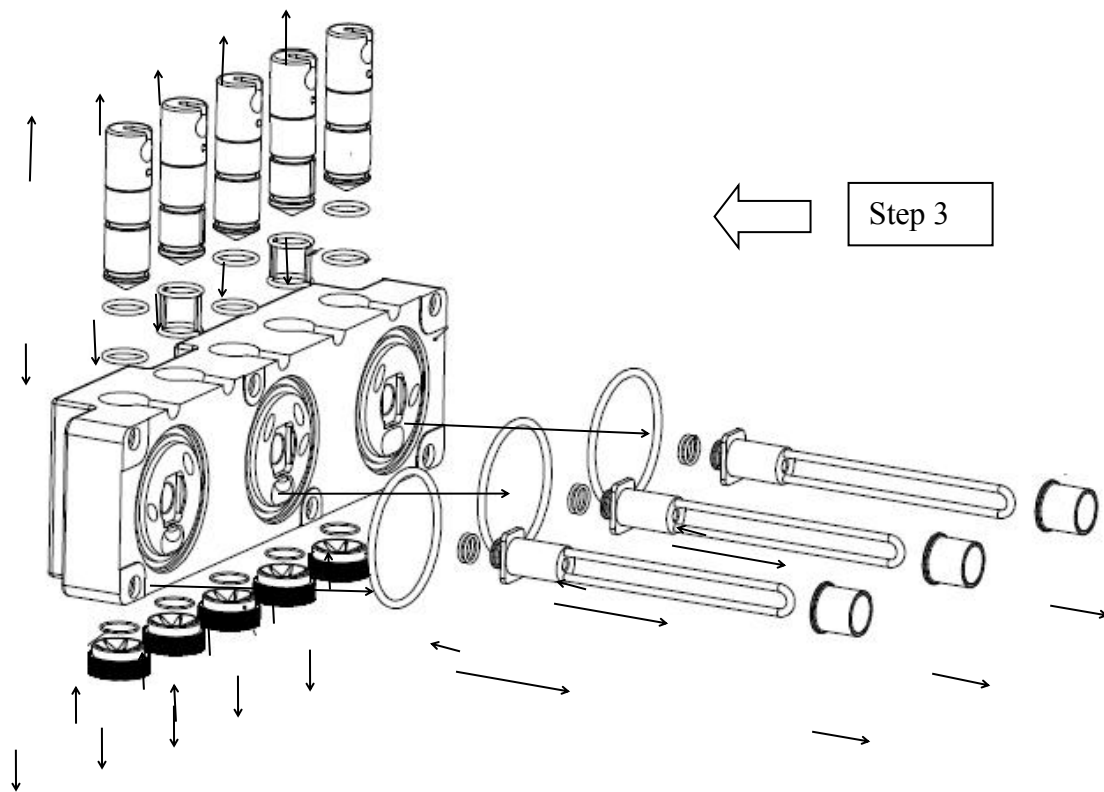
- Before you remove the neck block, please make sure that the entire liquid has flowed out of the funnel and cylinder (ice cream, water and disinfectant included).
- Please operate the machine according to the manual, or the accident happens easier.



Step 1



Step 2



Step 3

Concrete steps

Step 1: click on the screen to pull up 3 pistons and loosen 4 nuts at anti-clockwise direction.

Step 2: remove the nuts and the throat block component.

Step 3: remove the gasket on the throat block carefully.

Step 4: pull out the pistons.

Step 5: remove the gasket on the pistons.

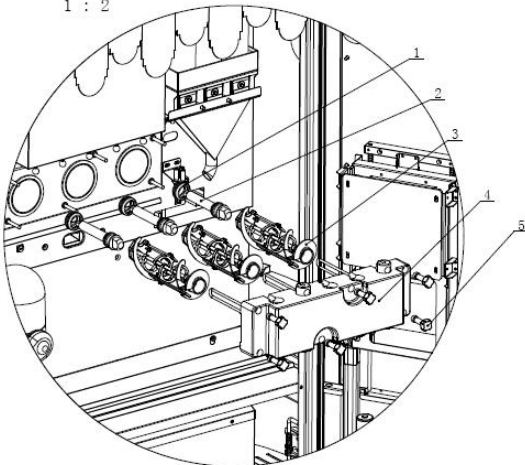
Note: 1) After cleaning, cover the lube on the gasket before installation.

2) If you need to install the throat block again, please follow the step 5 to step 2 (backwards).

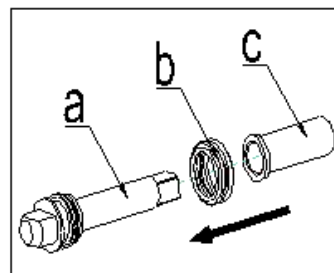
6. Stainless steel stirring, transmission shaft and sealing gasket

- 1) Before removing the stirring shaft, transmission shaft and gasket, you should stop the machine working.
- 2) Push the stirring shaft into the cylinder. Rolling the module left and right is to ensure that the stirring shaft is located at the transmission shaft position. If the installation is correct, the stirring shaft won't be out of the cylinder.

$\frac{A}{1:2}$

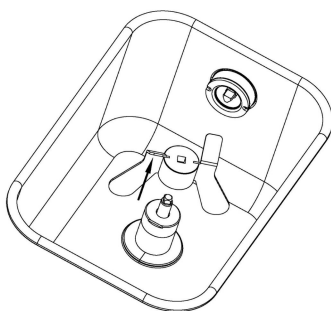


1. gasket for transmission shaft;
2. transmission shaft;
3. stirring shaft;
4. throat block;
5. bolt

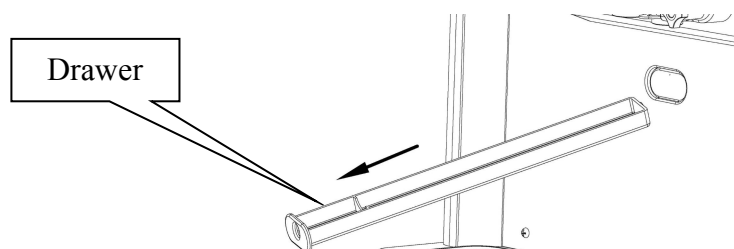


- a-transmission shaft;
b-gasket;
c-plastic sleeve for transmission shaft

- 3) Remove the mixing blade
Pull upward the mixing blade.



- 4) Pull out, clean and then install the drawer.



Note: If there is too much material in the drawer, you should replace the shaft seal or gasket.

Power connection

NOTE: The machine work in 220V, 50/60Hz. When the machine is used in 110V, a 8000W Watt 110V to 220V Electrical Power Voltage Transformer has to be used.

1. First plug in the transformer to the power source



2. Put the machine plug in the socket of transformer.



3. Turn on the transformer



Use the multi meter to measure power source on the electric box of machine, make sure the input electricity is 220V before turn on the machine power switch.

TROUBLESHOOTING GUIDE

Troubleshooting guide (1) :

No.	Malfunction	Cause	Solution method
1	Refrigeration becomes slower, compressor runs overtime.	1. Refrigerant is not enough. 2. Compressor bad exhausting. 3. Too much dust on the compressor with poor heat dissipation. 4. Excessive refrigeration	1. Inject refrigeration. 2. Repair or replace compressor. 3. Clean condenser. 4. Release refrigeration.

USER MANUAL

2	Compressor does not stop running.	1. Refrigeration part leakage. 2. Part ice jam. 3. Part dirt jam. 4. Compressor with low efficiency. 5. Hardness is setting high.	1. Refrigeration system vacuum pumping, and inject refrigerant. 2. Refrigeration system dry vacuum pumping, and inject refrigerant. 3. Check the jam position, clean or place capillary, expansion valve and filter with Nitrogen or Carbon tetrachloride. 4. Repair or replace compressor. 5. Lower the hardness setting.
3	Compressor is overheating.	1. Compressor lose oil. 2. Compressor runs overtime. 3. High room temperature. 4. Evaporator does not cooling with poor heat dissipation. 5. Part of coil for compressor is short circuit.	1. Add DH18 freezing oil. 2. Check and solve out the problem with Troubleshooting guide (2). 3. Change or improve working condition. 4. Repair refrigeration system, place the machine in a good ventilated placement to improve heat dissipation condition. 5. Replace compressor.
4	Compressor starts running, but sometimes does not run when switched on the machine.	1. Compressor has a little overheating with bad ventilation, so compressor starts running when in cooling, but heating protector is off and compressor stops running when in heating.	1. Improve heating dissipation condition. 2. Check bad connection wire and repair it.

Troubleshooting guide (2) :

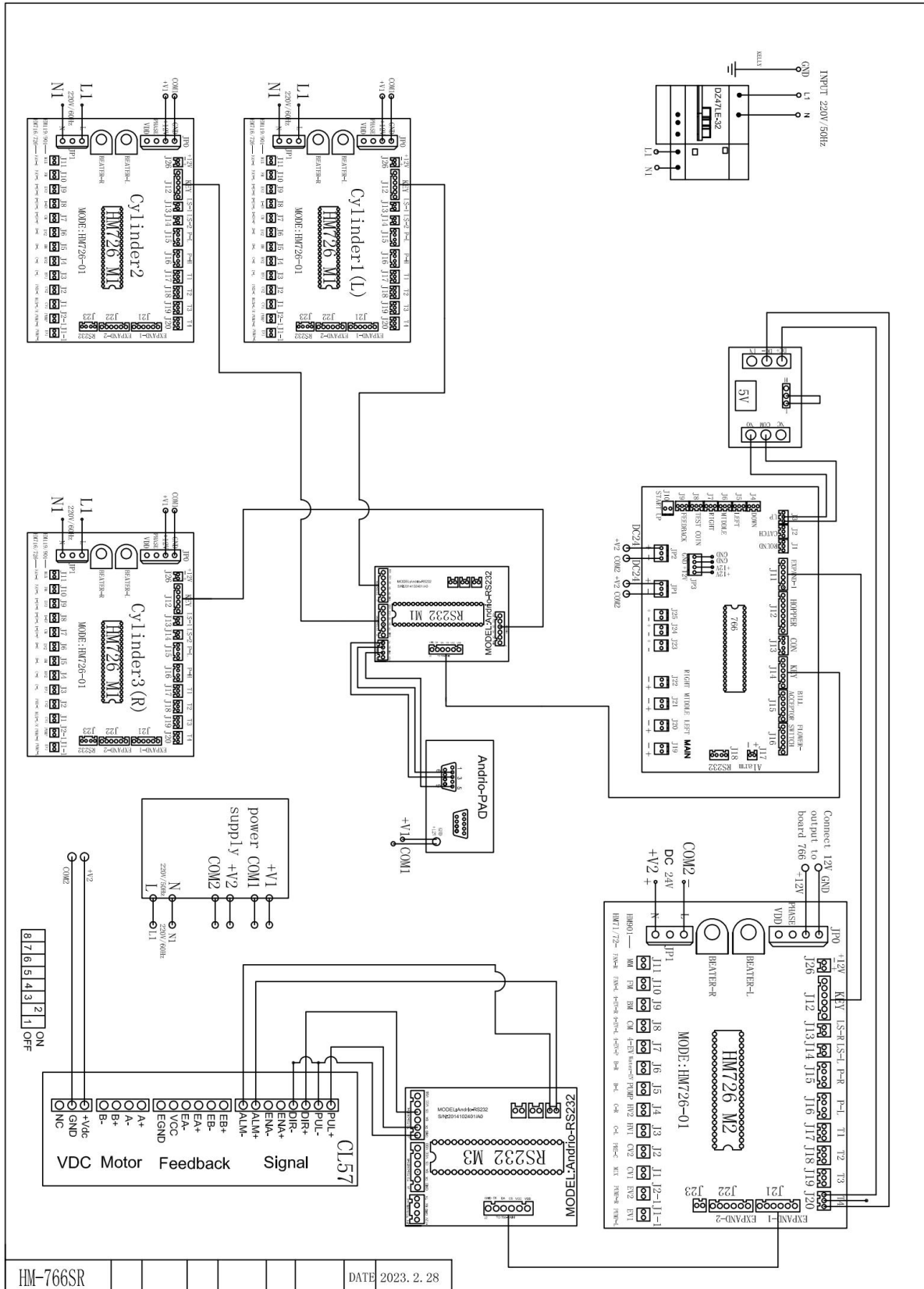
Malfunction	Cause	Analysis	Treatment
No cooling	Compressor does not run.	1. Compressor starter or capacitor is faulty. 2. Compressor is damaged. 3. The voltage becomes lower due to lower voltage of power supply and the smaller power line.	1/2. Replace faulty component. 3. Replace a big power line and increase the external power capacity.
	Ice jam or dirt jam	Refrigeration system with water or impurities.	Clean water and impurities in refrigeration system.
	Refrigerant leakage	Refrigeration system with bad sealing.	Repair the leakage position after check, vacuum pumping and inject new refrigerant.
Ice cream is not hard enough after finished.	Lower current setting value	1. Outside diameter of stirring shaft becomes bigger. 2. Main mother board current detection is error.	1. Reduce the outside diameter, or replace it. 2. Adjust the current value with the potentiometer on the main mother board.
Leakage (ice cream mixture or water)	Leaking from throat block	1. Damaged sealing rings or sealing gaskets on plugs or the throat block. 2. Bad installation for the fixing bolt.	1. Replace the sealing rings or the sealing gaskets. 2. Reinstall the throat block, and tighten it with the fixing bolts.
	Leaking from gear box	Oil sealed or sealing gasket in the gear box is damaged.	Replace it.
Ice cream	Little ice	1. Ice cream mixture jam the air	1. Remove and clean the expansion

USER MANUAL

is not dispensed out.	cream is dispensed out.	inlet, or air pump is damaged. 2. Little ice cream mixture in the hopper.	tube, or replace air pump. 2. Pour new ice cream mixture into hopper.
	Ice cream is too hard.	1. Little sugar in ice cream mixture. 2. Hardness is set lower.	1. Increase sugar quantity. 2. Adjust hardness setting.
	Stirring shaft does not run.	1. Motor or electric circuit is faulty. 2. Loose belt. 3. Faulty gear box.	1. Replace faulty component. 2. Adjust the motor position. 3. Repair or replace gear box.
	Ice jam on the throat block outlet.	In cleaning, water on throat block is not cleaned.	Remove the throat block and clean again.
No message displays on keyboard.		1. Fuse on the circuit board is faulty. 2. The circuit board is faulty.	1. Replace the fuse. 2. Replace the circuit board.

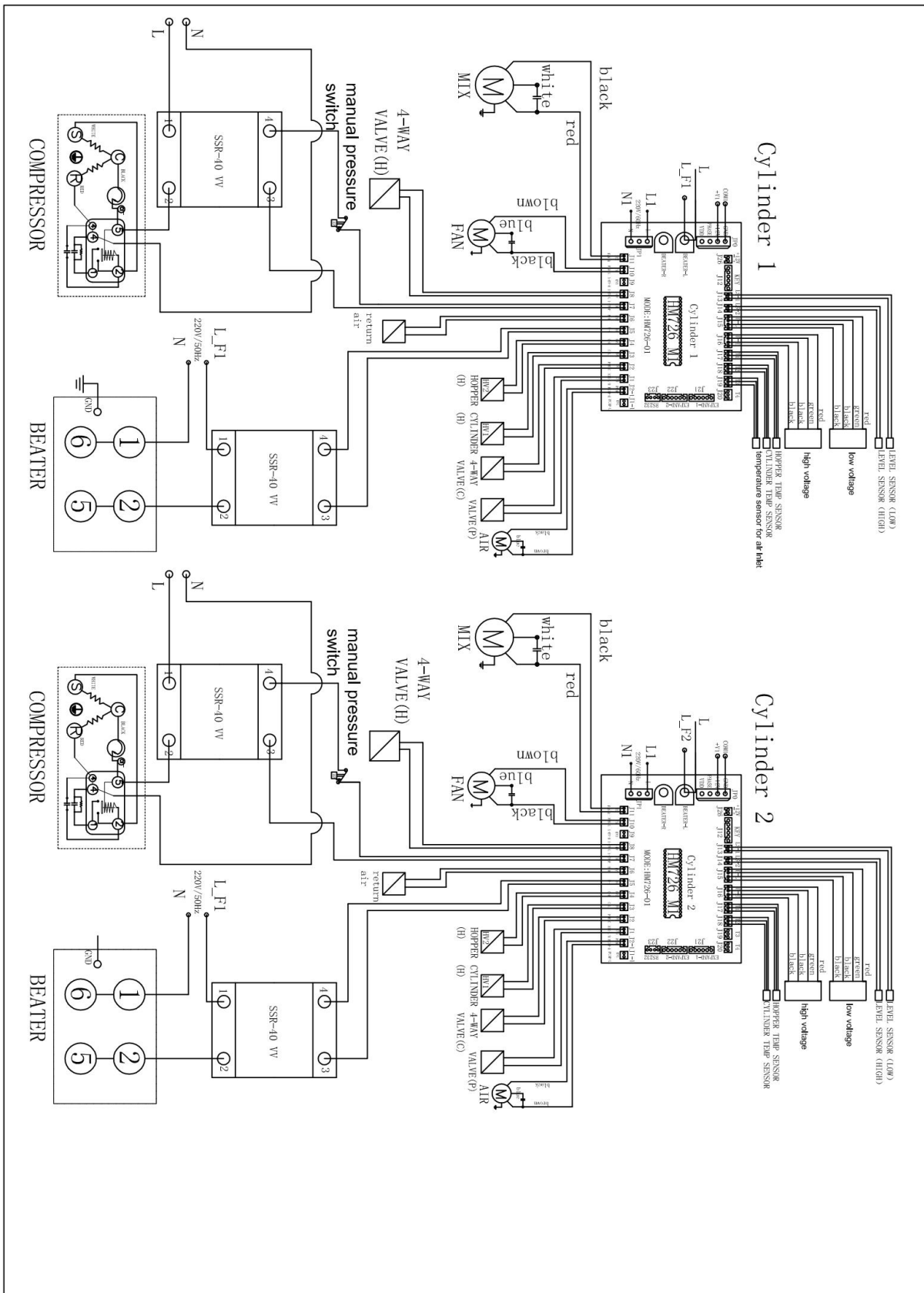
CIRCUIT WIRING DIAGRAM

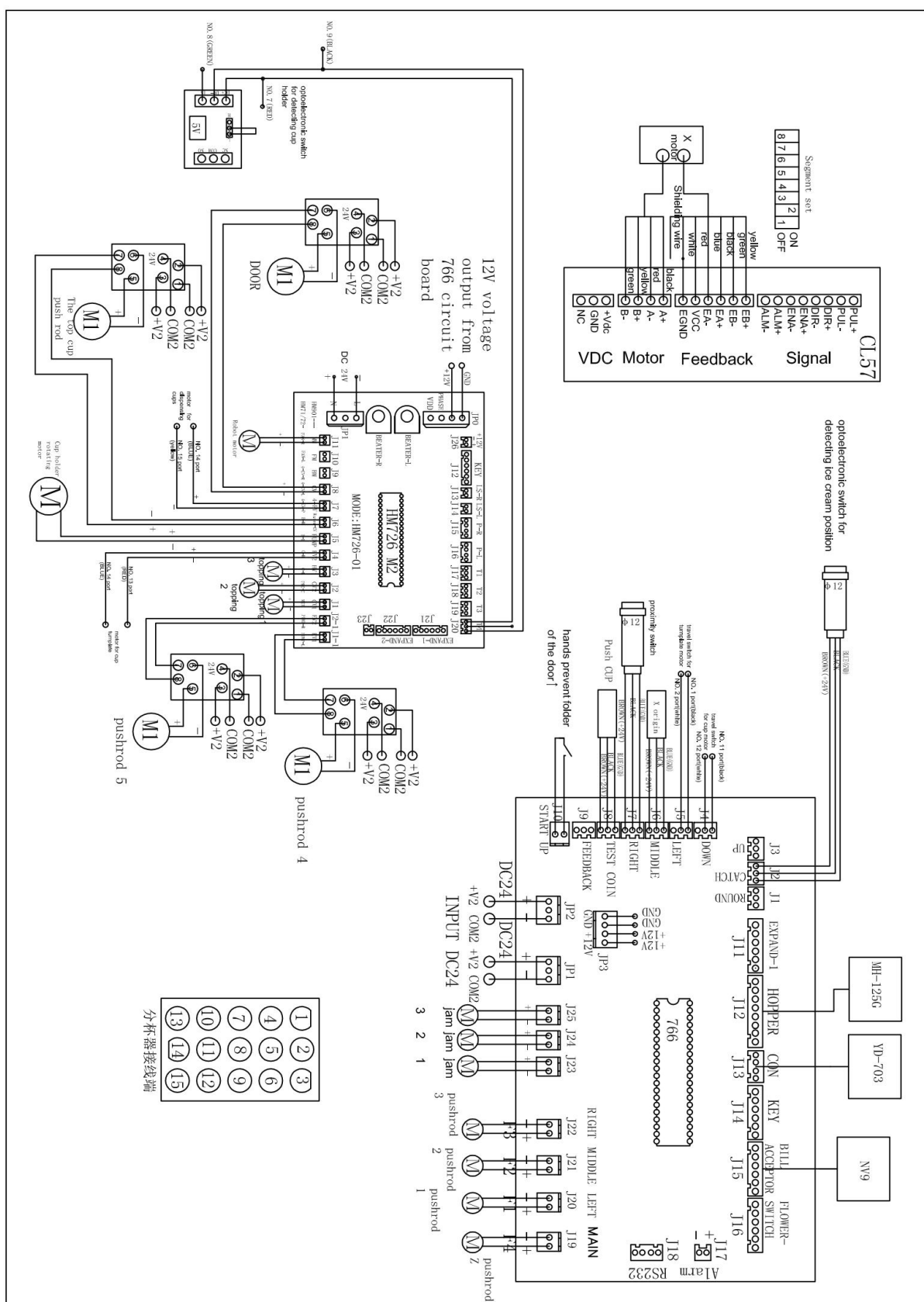
(for 220V, 50Hz)

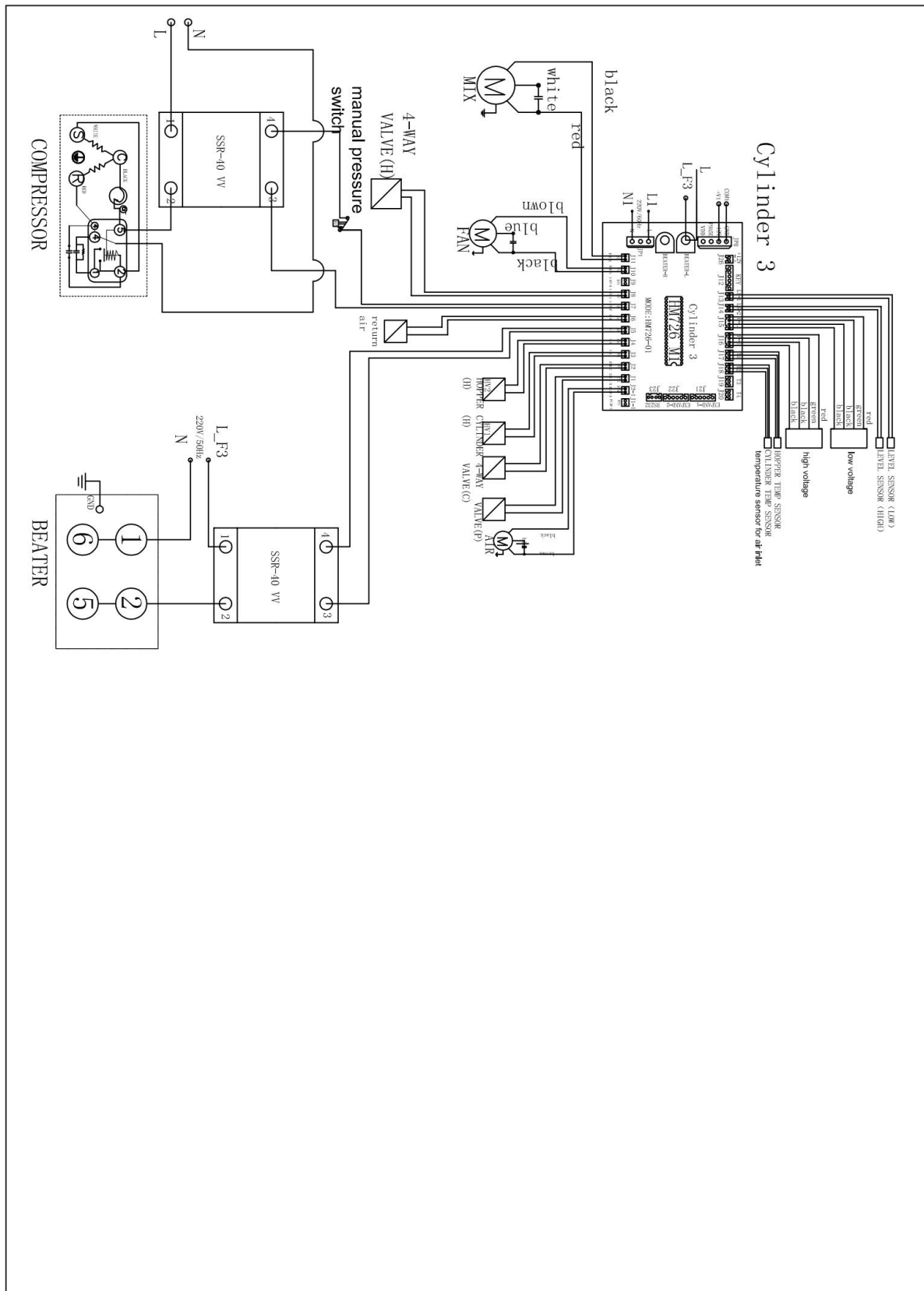


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DATE 2023. 2. 28







The company reserves the right to the improvement of all models, subject to change, without notice!